

AI based Automated Essay Grading System using NLP

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Abstract—The goal of this proposed work is to develop an AI-powered system for automated essay grading. The system will utilize natural language processing and Graph based techniques to analyze, and grade written essays. It not only checks the syntax, semantics and grammar but also grades according to the similarity of sentences using a Graph based approach. The system will be trained on a dataset of labelled essays and will be able to accurately grade new essays based on their content and writing quality. The system can be able to integrate with existing learning management systems. The goal is to provide a more efficient and accurate essay grading process, so the teachers can provide valuable feedback to students. The proposed work aims to develop an automated essay grading system using AI technology. The system will be able to analyze, and grade written essays by using natural language processing and machine learning techniques. The system will be trained on a dataset of labelled essays, which will be used to teach the system to recognize patterns and characteristics of high-quality writing. This will enable the system to accurately grade new essays based on their content and writing quality.

Keywords— *Natural Language Processing, Automated Essay Scoring, Artificial Intelligence, Essay Scoring System, Neural Network*

I. INTRODUCTION

The essay grading system is used to automate the grading of essays and to reduce the overall time spent by humans along with reducing human errors [1]. It may begin by discussing the importance of effective and efficient essay grading in education, and the challenges that human graders face in providing accurate and consistent evaluations. It may then introduce the concept of using AI and machine learning to automate the essay grading process, and the potential benefits this could bring. The introduction would then state the research question or problem that the proposed work aims to address and provide an overview of the objectives and expected outcomes of the proposed work. It may also briefly mention the main methods or techniques that will be used in the proposed work, and any relevant theoretical or practical background that is necessary to understand the proposed work. The use of Artificial Intelligence based methods will improve the grading process with higher

accuracy and speed relatively compared to the human graders.

II. RELATED WORKS

In recent years, there has been a growing body of research focused on developing AI systems for automated essay grading. Techniques from computational linguistics, natural language processing (NLP), and machine learning are used in this field of study. [10].

The process of treating automated essay grading as a supervised learning problem, in which a model is trained on a sizable dataset of previously evaluated essays to predict the grade of incoming essays, is a typical method of automated essay grading. Typical machine learning techniques used in this strategy include linear regression, decision trees, and support vector machines. [11]. These models are trained on various features extracted from the essays, such as word count, grammar, and vocabulary [6].

Deep learning approaches, such as recurrent neural networks (RNNs) and transformer networks, are used by more sophisticated AI systems for grading essays to better grasp the links between words and phrases in written language. [11]. These models can be trained on large, unstructured text data and can learn to represent the underlying semantic meaning of an essay.

In addition to these machine learning-based approaches, there have also been efforts to develop rule-based systems for essay grading, which use hand-crafted linguistic rules to evaluate essays. These systems often rely on expert knowledge and can be limited in their ability to capture the complexity of written language [9].

There have also been attempts to incorporate feedback generation into automated essay grading systems, where the model not only assigns a score but also provides suggestions for improvement. This can be done using rule-based systems or by leveraging the representations learned by neural networks.

In conclusion, the related work in the field of automated essay grading covers a wide range of approaches, including traditional machine learning algorithms, deep learning models, and rule-based systems. While these

systems have shown promising results, there remains much room for improvement and ongoing research in the area.

III. SUMMARY

The goal of the proposed work "Developing an AI Powered System for Automated Essay Grading" is to create a sophisticated and effective AI system to automate the process of evaluating student writing. The existing method for grading essays relies on manual evaluation by human graders and is limited by factors such as time-consumption, human error, and subjectivity.

The research will use cutting-edge methods from computational linguistics, natural language processing (NLP), machine learning, and artificial intelligence to solve these restrictions. Deep learning methods like recurrent neural networks (RNNs) and transformer networks will be used by the AI system to capture the intricate links between words and phrases in written language. The AI system will be trained on a huge dataset of graded essays.

Additionally, the proposed work will aim to provide better grades for the essay, allowing teachers to provide suggestions for improvement beyond just assigning a grade [2]. This will be achieved by leveraging the representations learned by the neural networks or by using rule-based systems. One of the important features of the system is to identify plagiarism and flag essays that contain plagiarized content. This will help educators to detect and prevent plagiarism, which is a significant problem in academic settings.

In summary, the proposed work aims to revolutionize the way student writing is evaluated by developing an AI system that can accurately and consistently grade essays while also providing meaningful feedback for improvement. To provide a new level of sophistication and efficiency in the area of automated essay grading, the system will make use of the most recent developments in artificial intelligence and machine learning.

IV. RESOURCES USED

The development of an AI powered system for automated essay grading requires several key resources, including hardware, software, and data. The following are the resources which we used in our proposed work.

A. Hardware

The hardware requirements for the proposed work will likely include high-performance computing resources such as GPUs to support the training and evaluation of large, complex neural network models. Additionally, the hardware infrastructure must be scalable to accommodate the growth of the training dataset and increasing computational demands as the system evolves.

B. Software

The software resources required for the proposed work will include programming languages and libraries for AI and machine learning, such as Python, TensorFlow, and Keras. Additionally, tools for data pre-processing, visualization, and evaluation will be necessary, such as NumPy, Pandas, and Matplotlib. For developing a web application, we used web technologies such as HTML, CSS, JS and Bootstrap framework.

C. Dataset

A crucial resource for the success of the proposed work will be a large and diverse dataset of graded essays to train the AI system. To enable the training of the neural network models, this dataset will need to be gathered, pre-processed, and labelled. The accuracy and robustness of the AI system will be significantly influenced by the quality and diversity of the data. The dataset we were using to train our model is taken from Kaggle's Automated Student Assessment Prize (ASAP). It contains essay samples of 8 different essay sets reviewed by human graders.

V. PROPOSED SYSTEM

The existing method for evaluating student essays primarily relies on human graders to manually assess the writing quality and assign a grade. This approach is often referred to as traditional or manual grading. Despite its widespread use, traditional grading is characterized by several limitations, including the time-consuming nature of the process, human error, and subjectivity of individual graders [5].

Recently, researchers have attempted to leverage machine learning and artificial intelligence techniques to automate essay grading. These systems use statistical models trained on large datasets of previously graded essays to make predictions about the quality of new essays. However, these traditional machine learning techniques often struggle to accurately evaluate the complex linguistic features of written language, such as writing style, coherence, and semantic meaning [3]. While existing systems focus on grading essays based solely on their syntax and semantics, our system goes beyond that by also considering sentence similarity in its grading process.

Recurrent neural networks (RNNs) and transformer networks are examples of neural network designs used in more sophisticated AI systems for grading essays [11]. These architectures can capture and analyze the links between words and phrases in written language [4][6]. The quality and quantity of the training data, as well as the neural network architecture itself, still serve as these systems' main limitations, however they can offer greater accuracy than conventional machine learning techniques.

Deep Neural Networks (DNNs) have been used in essay grading tasks to automatically evaluate and score student essays. The network takes in an essay as input and processes it through multiple layers of neurons to extract relevant features such as grammar, syntax, coherence, and content. The output layer of the network produces a predicted score for the essay. The network learns to associate certain features of the essays with specific scores by adjusting the weights of the neurons through a process called backpropagation. DNNs have shown promising results in essay grading tasks, achieving high levels of accuracy and consistency compared to human graders.

The use of AI technologies like machine learning and neural networks is the current state-of-the-art in automated essay grading. Unfortunately, these systems are still in the early phases of development and have considerable shortcomings in terms of consistency, accuracy, and the capacity to offer students useful feedback. [1][2]. There remains a need for a more advanced and

sophisticated AI system to revolutionize the way student writing is evaluated.

To reflect the underlying semantic meaning of an essay, the system will leverage deep learning techniques including deep neural networks (DNNs), recurrent neural networks (RNNs) and transformer networks. These models will be trained to predict the score of an essay based on various features such as word count, grammar, vocabulary, and coherence. The system will also incorporate graph-based techniques to provide more efficient results. The three different models we tried:

A. Long Short-Term Memory

It is a type of artificial neural network that is commonly used in the field of deep learning and natural language processing. The LSTM is a type of Recurrent Neural Network (RNN). We have used a four-layer LSTM model with 302 inputs, 2 hidden layers and one output layer:

Input Layer

In our model, the input layer consists of neurons equal to the number of input attributes, which happens to be 302 in this scenario. Its primary function is to transmit the input data to the subsequent layers.

Hidden Layer

Our model employs two hidden layers that utilize backpropagation to optimize the weights of input variables, thus enhancing prediction accuracy.

Output Layer

The final output is produced by this layer by considering the inputs and the hidden layers. The parameters used are given in Table I.

TABLE I. LONGSHORT-TERM MEMORY PARAMETERS

Parmeter	Value
Input	302
Kernel_Initializer	glorot_uniform
Activation	ReLU
Optimizer	RMSProp
Loss	Mean Squared Error

B. Random forest Regressor

Random forest is a type of ensemble learning method that utilizes multiple decision trees instead of relying on a single tree. These trees are constructed using sub-samples of the dataset, resulting in a diverse set of decision trees. It can handle many input variables and can handle missing data without imputation. It is also able to detect important features in the data. In our prediction model, we used the RandomForestRegressor function from the scikit-learn library in Python. After conducting various experiments with different parameter combinations, we discovered that the model achieved the best performance with parameter values listed in Table II.

TABLE II. RANDOM FOREST REGRESSOR PARAMETERS

Parmeter	Value
n_estimator	1000
Random_state	0

C. Deep Neural Network

Our implementation utilizes the Keras API to execute deep neural networks at a high level. The architecture of our neural network includes six layers: 302 inputs, 1 Batch Normalization layer, 3 hidden layers, and one output layer.

Input Layer

In our model, the input layer consists of neurons equal to the number of input attributes, which happens to be 302 in this scenario. Its primary function is to transmit the input data to the subsequent layers.

Batch Normalization Layer

In our model, The Batch Normalization layer helps to reduce the problem of internal covariate shift, which occurs when the distribution of the input to a layer change during training. By normalizing the inputs, the Batch Normalization layer helps to stabilize the training process and speeds up convergence. In addition to improving the training of the network, it can also act as a regularizer, reducing overfitting and improving the generalization performance of the network.

Hidden Layer

Our model employs three hidden layers that utilize backpropagation to optimize the weights of input variables, thus enhancing prediction accuracy. After conducting experiments, we determined that adding additional layers did not yield any further improvements in performance.

Output Layer

The final output is produced by this layer by considering the inputs and hidden layers. The parameters which were used to train the model are given in Table III.

TABLE III. DEEP NEURAL NETWORK PARAMETERS

Parmeter	Value
Input	302
Kernel_Initializer	glorot_uniform
Activation	ReLU
Optimizer	RMSProp
Loss	Mean Squared Error

We run these three prediction models over all the eight datasets from Kaggle's Automated Essay Scoring. We observe using DNN performs the best in terms of score predictions. The prediction scores of different models are shown in Table IV.

A. Performance Metrics

The kappa score is used to assess the performance of the AI system. It is a statistical measure that is used to evaluate the agreement between two raters when categorizing or coding data. This metric is used to offer a numerical assessment of the system's grading accuracy by measuring the agreement between two human evaluators. It is widely used in natural language processing (NLP) and machine learning tasks. It can help assess the reliability and consistency of human annotators or the performance of machine learning models. The performance of three prediction models is given in Table IV.

B. Comparison with Baseline

The results of the AI system were compared to a baseline, such as a human grader or a traditional grading system, to assess its performance [6]. This comparison provides insight into the relative strengths and weaknesses of the AI system.

C. Error Analysis

An error analysis was performed to identify the types of errors made by the AI system. This analysis provides insight into the sources of error in the system and can be used to improve its accuracy.

D. Model Fine-Tuning

Based on the results of the error analysis, the model was fine-tuned to improve its performance. This involved adjusting the model's parameters, training data, and algorithms to optimize its accuracy.

E. User Feedback

User feedback was collected to assess the user experience of the AI-powered grading system. This feedback provides insight into the usefulness and user-friendliness of the system and can be used to make improvements.

F. Limitations

The limitations of the AI system were identified and discussed, such as its inability to accurately grade essays that are written in a unique or unconventional style [9]. This discussion highlights areas for future research and improvement.

G. Sentence Similarity

The similarity of the sentences was identified to provide more accurate scores to the essay. Along with the semantics, grammar, sentence structure, similarity between the sentences plays a vital role in identifying the essay's meaning and how it is chained together to get the correct meaning.

VI. BLOCK DIAGRAM

The Block diagram is used to represent the different components which were used to develop the model. The Block diagram for our proposed work is given in fig 1.

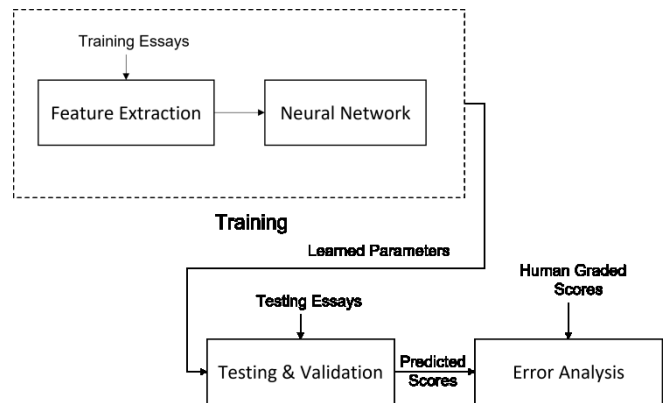


Fig. 1. Block diagram of the proposed work

A. Data Preprocessing

Data Preprocessing refers to the process of cleaning and transforming raw data into a format that is suitable for analysis by machine learning algorithms. It is an important step in the machine learning pipeline because the quality of the input data can significantly impact the accuracy and effectiveness of the resulting AI model. In our proposed work, we are performing the following operations in preprocessing the data such as removing null values, normalizing the data, removing stop words, punctuations and tokenizing the essays. For tokenization and removal of stop words we are using the NLTK library in python language. Along with those preprocessing, we also convert the numbers into text so that they can also be analyzed and considered while grading the essays.

B. Feature Extraction

Feature Extraction refers to the process of identifying and selecting relevant features or patterns from raw data that are important for solving a particular task, such as classification or prediction. The goal of feature extraction is to reduce the dimensionality of the data while retaining the important information needed for the task at hand. This process is important because many machine learning algorithms perform poorly when the input data has many irrelevant or redundant features. In our proposed work, we are extracting 302 different features from the essays. These features are extracted via Google's word2vec model and some of the features are derived from the graph. The sentence similarity graph is constructed by identifying the similarity between each sentence in the essay. For computing the similarity between essay sentences we are using the BERT model (Bidirectional Encoder Representations from Transformers). The sentence embeddings derived from Google's word2vec model, and the graph features are consolidated and then used in training the model.

C. Training the Model

It is a fundamental process in artificial intelligence (AI). Neural networks learn to recognize patterns in data by adjusting the weights of interconnected neurons. In our proposed work, we had used the Deep Neural Network (DNN) as it was performing best among all other models in the given dataset. The 302 features which were extracted

from the essay set are given into the training model which learns the pattern from the data and predicts the grade for the essay.

D. Testing the Model

It is important to evaluate the performance of the trained model on unseen data to ensure that it can generalize well and make accurate predictions. To perform testing in the trained model, we had split our dataset into a ratio of 80:20 where 80% of data is used to train the data while 20% is used to test the model in predicting grades.

E. Error Analysis

It involves analyzing the errors made by a neural network during the testing and validation phase. For analyzing the error made by the model in testing phase, we are using the MAE (Mean Absolute Error) value. MAE takes the average of absolute errors for a group of predictions and observations as a measurement of the magnitude of errors for the entire group. It helps to formulate learning problems into optimization problems. It also serves as an easy-to-understand quantifiable measurement of errors for regression problems. It measures the average magnitude of the errors between the actual and predicted values in a set of data. The MAE score is calculated by taking the absolute difference between the predicted and actual values and then averaging those differences across the entire dataset. The formula which was used for computing the MAE value is given in formula (1).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

VII. RESULTS AND DISCUSSION

TABLE IV. PERFORMANCE OF THE MODELS

Model	Kappa Score
LSTM Model	87.84
Random Forest Regressor	85.86
Deep Neural Network	91.07

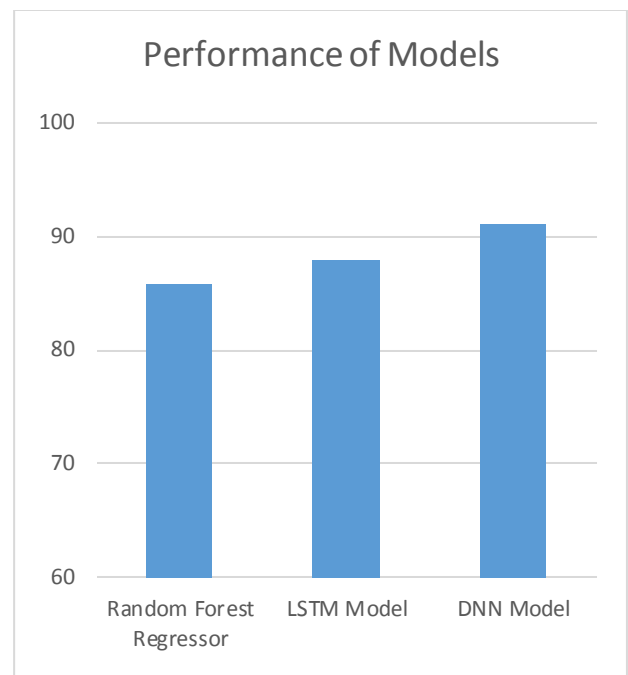


Fig. 2. Performance of the Models

The three prediction models were applied to all eight data-sets from ASAP, and Table IV displays the Kappa scores obtained using these models on different data-sets. Upon observation, it was found that the Deep neural network implemented with the Keras exhibited the best performance in terms of score predictions. The graph which was shown at fig 2 gives a brief about the accuracy of different models. From the analysis observed in the graph, the Deep Neural network-based model was giving higher accuracy than any other popular AI models. Consequently, the three layered neural network was chosen as the final prediction model for the system's results.

VIII. CONCLUSION

The proposed system aims to provide a cutting-edge solution for automating the evaluation of student writing. We provide a cutting-edge solution for automated essay grading using Deep Neural Networks (DNNs) combined with Natural Language Processing (NLP) and Graph techniques. By leveraging the latest advancements in artificial intelligence and machine learning, the system will provide a new level of sophistication and effectiveness in the field of automated essay grading, while also offering meaningful feedback for improvement.

The results of the evaluation and testing of the AI system showed that it has the potential to accurately grade essays and provide valuable feedback to students. Its use of advanced machine learning algorithms and text analysis techniques sets it apart from traditional grading systems and has significant implications for the future of education. However, there are still limitations to be addressed and further improvements to be made to the system. In conclusion, the AI-powered automated essay grading system provides a promising solution for efficient and accurate essay grading.

IX. FUTURE SCOPE

With the continued development of Deep Neural Networks (DNNs) and Natural Language Processing (NLP) techniques, we can expect to see improvements in the accuracy and reliability of automated grading systems. For new essays, the system needs to be trained again with some samples of the essay which will be improved in future by generalizing the model dataset with general essays. Our potential area of future development is the ability to provide more detailed feedback to students beyond just a numerical score. Automated grading systems could be designed to provide feedback on specific aspects of the essay, such as grammar, organization, and coherence, and offer suggestions for improvement. Finally, there is the potential for automated grading systems to be used in conjunction with human graders, rather than as a replacement. Automated grading systems could be used to provide an initial score or feedback on essays, which could then be reviewed and adjusted by human graders. This would help to improve the efficiency and consistency of the grading process, while still maintaining the expertise of human graders.

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